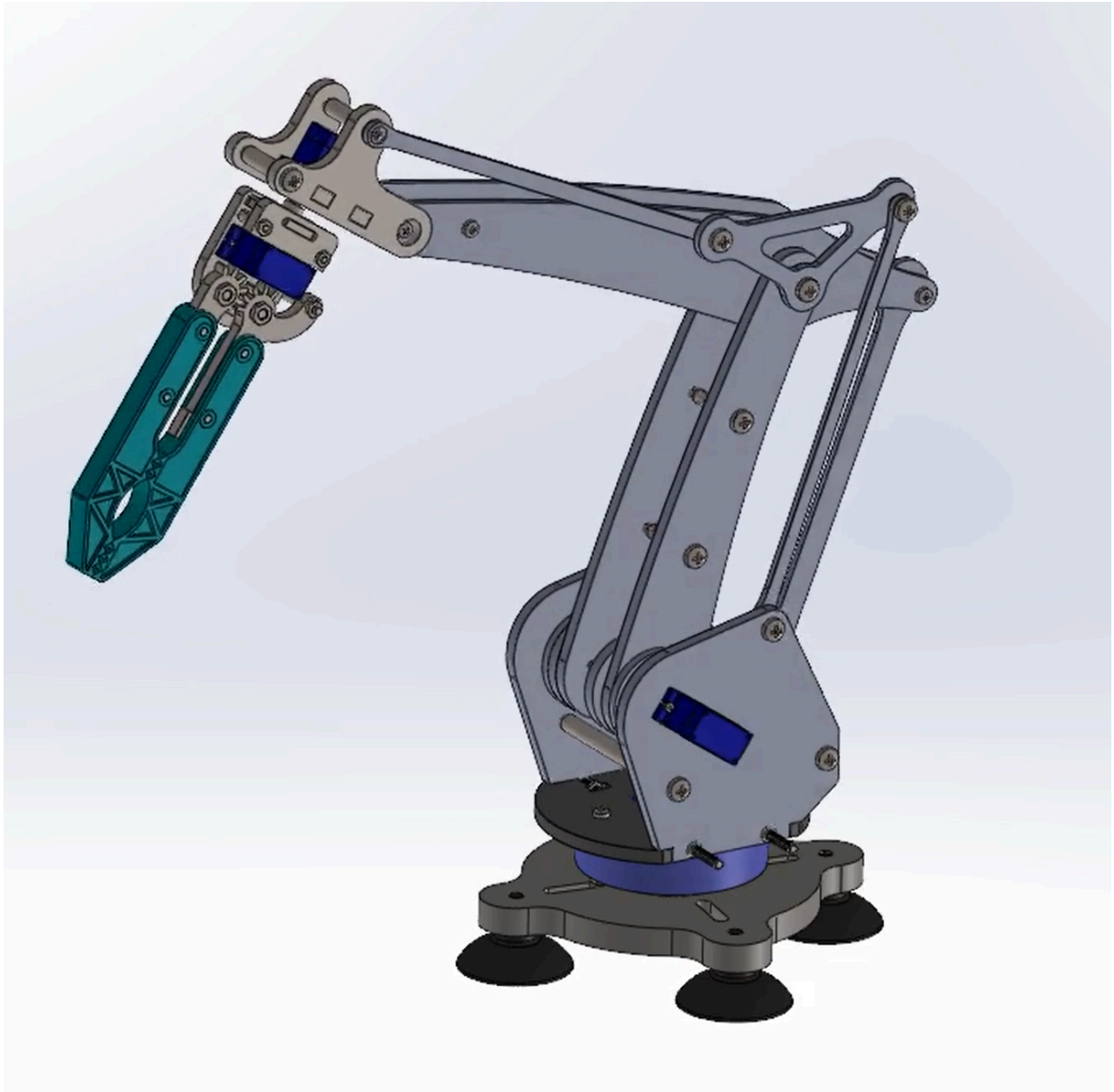


Internship Program 2026

CodeAlpha

Robotic Arm Design



4-DOF PARALLEL-LINK ROBOTIC ARM DESIGN

Base Rotation (Axis 1 - Yaw): A revolute joint rotating parallel to the desktop surface, driven by a servo housed inside the circular blue base ring.

Shoulder Pitch (Axis 2): A lower revolute joint driving the main structural upright beam forward and backward.

Elbow Pitch (Axis 3): A remote revolute joint actuated via a **rear parallel-link tie rod**. This rod allows the servo mounted at the base chassis to pull the upper forearm triangular link up and down, keeping the motor weight off the moving upper arm.

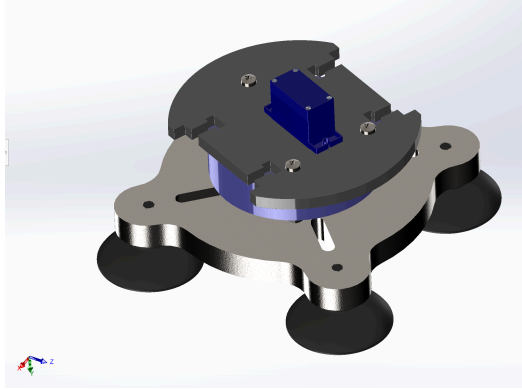
End-Effector Actuation (Axis 4 - Grip): A single micro-servo integrated into the wrist bracket that drives a synchronized spur gear mechanism to close the teal dual-finger claw.

COMPONENT PIECE-BY-PIECE BREAKDOWN

H-Shaped Ground Anchor Base: A flat, four-cornered base plate fitted with **four soft rubber suction cups** to firmly clamp the robot to a desk without permanent hardware screws.

Actuation Housing Hub: The circular blue interface ring that contains the internal gearing or servo horn interface for 360-degree or limited-angle horizontal rotation.

Dual-Plate Main Chassis: The lower gray structural block made of two parallel plates held together by structural pins. It holds the main **blue micro-servo motor** responsible for operating the rear linkages.



Parallel-Link Mechanical Boom: Consists of a primary thick center arm and a secondary thin rear link rod. These attach to a triangular pivot bracket at the top, ensuring the gripper maintains a stable angle relative to the ground during motion.

Teal Mechanical Gripper: A compact wrist unit carrying a dedicated blue micro-servo. The servo shaft links directly to integrated gears on the roots of the two teal claw jaws, converting rotational servo motion into symmetric clamping force.

